

test drive ■ portable storage

on the side of the device, and it was difficult to even activate the switch—we finally succeeded, using a pin. Some drives such as the ones from Transcend and the Freecom FM-1, were also bundled with password protection utilities.

Speaking of utilities, all the drives, except the ones from Princeton, had some software and other accessories bundled. The Princeton drives were the only ones to be delivered without any accessories whatsoever—no driver CD, or USB cable. Naturally, they scored the lowest in our features comparison, and finished at the bottom. The Nexdisk PSD-64N had possibly the most valuable tool bundled along, Jungsoft's DataSheriff backup and recovery software. It also had the standard USB cable, a neck-strap and even a quick start guide, as did most other drives.

The TS512MJFLASHA won the Best Value award due to its above-average performance and low price of Rs 9,000



Performance

If thumb drives are to be utilised extensively on the move, they need to be fast and perform well. The Sisoft Sandra and HDtach benchmarks helped us assess the theoretical speeds and access times of the drives, to find and open any random file located on it.

Sisoft Sandra Professional 2003: The drive index is a combined score indicating the overall performance of the device, and it represents its performance in everyday usage. The maximum drive index value we obtained was from the Princeton 64 MB drive, followed closely by the Touchmate model, with scores of 945 and 944 respectively. The 128 MB Princeton drive followed in a close third, with a drive index of 941. Although nearly all the drives managed to perform well, the Freecom FM-1 and the Nexdisk PSD-64N finished at the bottom with their

drive indices not even crossing the 700 KBps mark. The low drive index scores means that they will be a lot slower than the others to copy and move larger files such as video and audio clips. The results were similar for the random write speeds: the Nexdisk 64 MB drive logged the lowest score of 360 KBps, while the Touchmate and the Princeton drives were the fastest again.

HDtach 2.7: The read burst speed for all the drives remained constant at 1 MBps. Similarly, the CPU usage of the drives remained negligible—only the Transcend 512 MB and Princeton 512 MB logged values of 1.2 and 1 per cent respectively. This low CPU utilisation is ideal for laptops, and will leave the processor free to tackle other more important system tasks. The random access times also hovered around 6 milliseconds across all the drives—the exceptions being Nexdisk and Transcend again. In fact, the Nexdisk showed a pathetic random access time of 68.5 milliseconds.

Real-world tests: Unlike synthetic benchmarks, real-world tests measure the actual performance of the device in normal usage conditions. In the first test, the Princeton 128 MB drive completed the assorted file read test with the fastest time of 23.45 seconds. Coming in second was its 64 MB sibling, with the Easy-Disk and the Touchmate 256 MB drives following. The assorted write test had nearly the same results, except that the Princeton 64 MB drive topped this time. The Freecom FM-1 stood dead last, with a bad time of 117 seconds—about double that of the other times. The assorted file tests measured the time taken to transfer a random collection of data such as a set of documents, presentations, images, etc., which is what a user would normally transfer. Thus, low scores in this test indicate a drive's inability to handle varying amounts of data while copying and writing.

Since the storage capacities varied substantially, we decided to carry out one more test: we checked each drive's transfer speeds when handling their maximum capacity of data. As the drive's maximum capacities were varied, we created separate assorted test folders and sequential data files for each capacity. We divided the amount of data transferred (MB) by the time taken for the transfer (s) to come up with the real data transfer rate (MBps).

The Princeton 128 MB drive took top spot with a sequential transfer rate of 2.3 MBps. The Transcend 128 MB came second with a transfer rate of 2.06 MBps, while the 128 MB Apacer Handy Drive finished last with 0.93 MBps. In the assorted file transfer test, none of the drives could cross the 1 MBps mark; the 256 MB Princeton came closest with a speed of 0.85 MBps.

Value for money

The one real advantage with USB flash drives is that they have absolutely zero maintenance and a long life span—the specifications in the manuals claim a lifespan of about 10 years. However, deciding on the most cost-efficient drive is also important, as buying the largest drive doesn't always guarantee you maximum benefits. This is where our value for money quotient helps ascertain the right model for you.

In terms of actual value for money, the Transcend 512 MB model is quite a steal—it costs only Rs 9,000 and performed well in our tests. The other 512 MB drive, the Princeton MPD512, costs four and a half thousand more. The most uneconomical drive in our comparison was the Freecom FM-1 drive, which has a whopping cost of Rs 72.58 per MB of storage space.

Princeton 1 GB USB 2.0

In the thumb drives shootout, we actually received 15 pieces for carrying out our tests, but had to leave the Princeton 1 GB USB 2.0 drive out of the comparison. This was only due to one reason—with its faster interface, this drive simply blew the competition away. For example, in the Sisoft tests, the highest drive index score of 945 was reported by the 64 MB Princeton model. The 1 GB Princeton drive obliterated this score, recording a drive index of 7,350—over seven and a half times higher. With such a lop-sided comparison, we had no choice but to exclude this drive from the main competition. This drive also showed random

read and write speeds of 8 and 2.2 MBps respectively—comparable to some of the values reported in the external hard drives comparison.

It was the same story in the HDtach benchmark: the 1 GB Princeton logged the fastest random access time in the entire comparison—fastest in both the thumb drives and external hard disk section. In the real world maximum capacity tests, the sequential read and write speeds were an astounding 6.20 and 6.55 MB/sec respectively. It is somewhat expensive with a price tag of 28,000 and a cost per MB of a little under Rs 28. With its superlative performance and high capacity, this is the best you can get—if you can afford it.

